Association Rule Mining — Terminology and Core Measures

Basics of Data Mining — Week 9

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1. Where We Are in the Course

In this course we are studying **unsupervised** ways of finding patterns in data. There is no target variable and no "right answer". The question is simply: *given some data, what is interesting, and what is not?* The aim is to turn that information into something **actionable**.

So far we have covered:

- **Data reduction:** DFT and PCA
- **Clustering:** k-means

We now begin **Association Rules**, or **Association Rule Mining (ARM)**. It is also unsupervised: given a dataset, find the interesting relationships inside it.

The mathematics is gentle. We count things, convert the counts to percentages or probabilities, and build intuition from them. The only real difficulty is keeping the **terminology** straight, so we introduce each term through a single running example.

> **Note for Review** Every quantity in this lecture (support, confidence, lift) is just a **count divided by a count**. If you can read the definition carefully and decide *what to count* and *what to divide by*, you can compute anything here.

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2. The Running Example: A Streaming Service

Suppose we observe **viewing sessions** on a streaming service. The sessions may come from different viewers or from the same viewer at different times. In each session we record which content was watched. The questions we care about are of the form: *if someone watched A, did they also watch B?*

The idea is general. A store with 20 products where customers buy A and then also buy B is the same problem. So is a website where we record which pages a visitor clicked.

How to read it:

- In session T_4 the viewer watched A and also D.
- In session T_3 the viewer watched B and also E.
- T_7 is identical to T_1. This is allowed, because two different people can watch exactly the same things.

The data we will use throughout:

Transaction (Session)	Items (Watched)
T_1	A, B, C
T_2	A, B
T_3	B, E
T_4	A, D
T_5	A, B, D
T_6	D, E
T_7	A, B, C
T_8	B, D

3. Items, Transactions and the Database

3.1 Item

An **item** is the most fundamental (elementary) element of a transaction. In our example the smallest unit is a piece of content. A viewer either watches content A or does not, and nothing is smaller than that. So each of A, B, C, D, E is an item.

There are **5 items** but **8 transactions**.

3.2 Item Universe

Putting all available items together in one set gives the **item universe**:

$$I = \{A, B, C, D, E\}$$

3.3 Transaction

A **transaction** is the *observed collection of items* in one session (one purchase, one visit, one viewing session). For example, $T_5 = \{A, B, D\}$ is one transaction containing three items. The viewer watched them one after another, but they are grouped together as a single unit.

Every transaction is a subset of the universe, because it can only contain items that exist:

$$T_j \subseteq I \quad \text{for every } j$$

3.4 Transaction Database

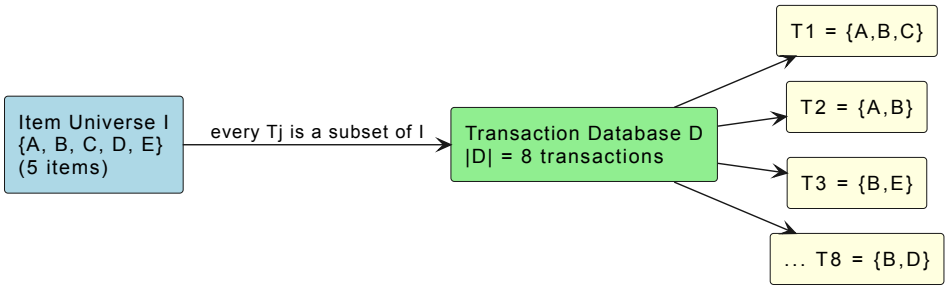
The set of all transactions is the **transaction database**:

$$D = \{T_1, T_2, \dots, T_8\}$$

3.5 Cardinality of D

$$|D| = 8$$

$|D|$ is the **number of transactions**, not the number of items inside them. Even though T_1 holds three items, it counts as **one** transaction.



> **Note for Review** Three words, three levels. **Item** is one object. **Transaction** is the group of items that appeared together *in the data*. **Database** is the collection of all transactions. Mixing up "number of items" with "number of transactions" is the most common slip in this chapter.

4. Itemsets

An **itemset** is any set built from items of the universe. It can contain one item, two, three, or more. An itemset is a subset of I.

Itemset versus transaction. A transaction is also an itemset. For instance $T_2 = \{A, B\}$ is a set of items. The difference lies in *who chose it*:

- A **transaction** was chosen by the viewer. It is the **data**.
- An **itemset** can be defined by **us** from any of the available items. We could write $\{A, D, E\}$ even if no viewer ever watched exactly that combination.

Example	Name
$X = \{A\}$	1-itemset
$X = \{A, B\}$	2-itemset
$Y = \{A, B, D\}$	3-itemset

5. Occurrence, Support Count and Support

Now we start computing. These steps are only counting.

5.1 Occurrence

Take the itemset $X = \{A, B\}$. We say X **occurs in** a transaction if *both* A and B are present in it. Checking the table:

Transaction	Contains A and B?
$T_1 = \{A, B, C\}$	Yes
$T_2 = \{A, B\}$	Yes
$T_3 = \{B, E\}$	No (no A)
$T_4 = \{A, D\}$	No (no B)
$T_5 = \{A, B, D\}$	Yes
$T_6 = \{D, E\}$	No
$T_7 = \{A, B, C\}$	Yes
$T_8 = \{B, D\}$	No (no A)

So X occurs in T_1, T_2, T_5, T_7. These are four places where the two items occur **together**.

5.2 Support Count

The **support count** of an itemset is the number of transactions that contain it:

$$\sigma(X) = 4 \quad \text{for } X = \{A, B\}$$

The support count belongs to the **itemset as a whole**, not to A alone or B alone. If the itemset were $\{A, B, C\}$ we would count only the transactions where all three occur together.

5.3 Support

The **support** is the support count divided by the number of transactions:

$$\text{supp}(X) = \frac{\sigma(X)}{|D|}$$

For $X = \{A, B\}$:

$$\text{supp}(\{A, B\}) = \frac{\sigma(X)}{|D|} = \frac{4}{8} = 0.5 = 50\%$$

Interpretation: 50% of the observed viewing sessions contained both A and B. In other words, half of the sessions had A and B together.

5.4 Support as a Population-Level Quantity

When we say $\text{supp}(A, B) = 50\%$ we are describing the **population** of sessions, and we treat the observed sessions as the population here. In statistics, a **sample** is a subset we observe, and a **population** is the whole group we want to describe. Support is a population-level measure.

> **Note for Review** Quick counts for the other single items, since they are needed later: $\sigma(A) = 5$, $\sigma(B) = 6$, $\sigma(C) = 2$, $\sigma(D) = 4$, $\sigma(E) = 2$. Dividing each by 8 gives the supports $0.625,\ 0.75,\ 0.25,\ 0.5,\ 0.25$.

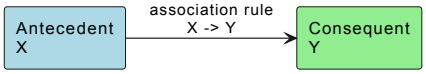
6. Association Rules

An **association rule** has the form

$$X \to Y$$

It has two parts:

- **Antecedent:** the left-hand side, X
- **Consequent:** the right-hand side, Y



6.1 What the Arrow Means

Read $A \to B$ as: *what is the probability that B was viewed, given that A was viewed?*

$$A \to B \ \text{ means } \ P(B \mid A)$$

This is a **directional and conditional relationship**:

- **Directional:** $A \to B$ and $B \to A$ are two *different* rules.
- **Conditional:** it is defined through a conditional probability, and in general

$$P(B \mid A) \neq P(A \mid B)$$

> **Tip:** Whenever you see an arrow like this, think *probability*. That removes most of the confusion. Everything here comes from counting and dividing, which is the frequentist way of estimating a probability.

6.2 X and Y Can Be Itemsets

Neither side has to be a single item. For example:

$$\{A, B\} \to \{C, E\}$$

This asks: *if somebody watched A and B together, what is the probability that they also watch C and E?* Both X and Y are just itemsets. The arrow keeps its direction and its conditional meaning.

7. Confidence

Confidence measures how often the **consequent** appears among the transactions that contain the **antecedent**.

$$\text{conf}(X \to Y) = P(Y \mid X)$$

In counts:

$$\text{conf}(X \to Y) = \frac{\sigma(X \cup Y)}{\sigma(X)}$$

7.1 Worked Example: $\text{conf}(A \to B)$

Step 1. Count the antecedent. Where does A occur?

$$A \text{ occurs in } T_1, T_2, T_4, T_5, T_7 \;\Rightarrow\; \sigma(A) = 5$$

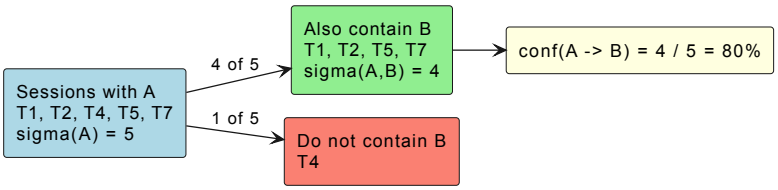
Step 2. Count antecedent and consequent together. Where do A and B occur together?

$$\{A, B\} \text{ occurs in } T_1, T_2, T_5, T_7 \;\Rightarrow\; \sigma(A, B) = 4$$

Step 3. Divide.

$$\text{conf}(A \to B) = \frac{\sigma(A, B)}{\sigma(A)} = \frac{4}{5} = 0.8 = 80\%$$

Interpretation: among the sessions containing A, 80% also contained B.



7.2 Important Remarks

1. **These are actual numbers.** Every quantity so far (support, confidence, lift) is computed directly from the given dataset. Nothing is predicted. This is data-based analysis, not predictive analysis.
2. **No causality.** A confidence of 80% for $A \to B$ does **not** say that watching A *caused* the viewer to watch B. It only says that people who watched A also tended to watch B. Causality is a different topic that we are not entering.
3. **Order of counting matters.** Count the *antecedent* first, because it is the denominator. Then count the *antecedent and consequent together*.

> **Note for Review** Confidence has a **direction**. $\text{conf}(B \to A) = \sigma(A, B)/\sigma(B) = 4/6 \approx 0.667$, which is not the same as $\text{conf}(A \to B) = 0.8$. The numerator is the same, but the denominators differ.

8. Lift

Confidence alone can be misleading. A consequent that is very common will show high confidence with almost *any* antecedent. Lift corrects for this by comparing the rule against the **baseline popularity of the consequent**.

8.1 Support of B

$$B \text{ occurs in } T_1, T_2, T_3, T_5, T_7, T_8 \;\Rightarrow\; \sigma(B) = 6$$

$$\text{supp}(B) = \frac{\sigma(B)}{|D|} = \frac{6}{8} = 0.75$$

8.2 Definition of Lift

$$\text{lift}(X \to Y) = \frac{P(Y \mid X)}{P(Y)} = \frac{\text{conf}(X \to Y)}{\text{supp}(Y)}$$

This is a matter of definition. It divides the probability of Y *given* X by the probability of Y *on its own*.

8.3 Worked Example: $\text{lift}(A \to B)$

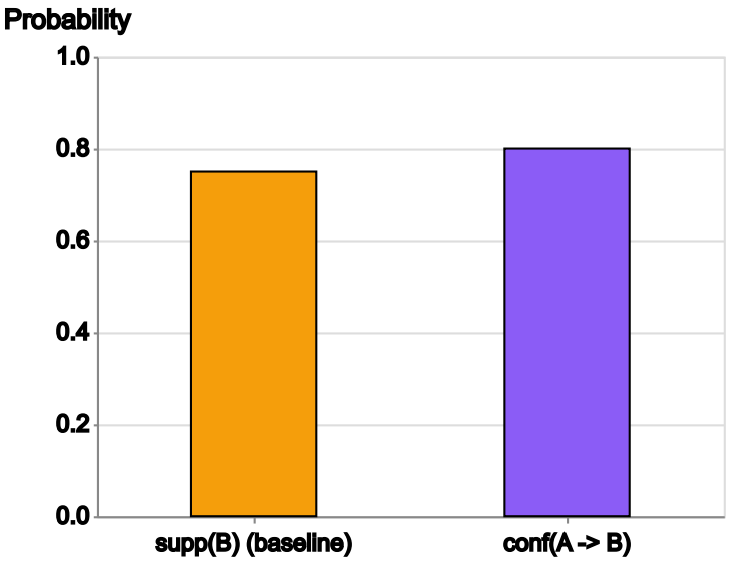
$$\text{lift}(A \to B) = \frac{\text{conf}(A \to B)}{\text{supp}(B)} = \frac{0.8}{0.75} = 1.0667 \approx 1.07$$

8.4 Interpreting Lift

A lift of 1 is the **baseline**. We always compare against it.

Lift value	Meaning
$\text{lift} > 1$	**Positive** association
$\text{lift} = 1$	No association (the baseline: X tells us nothing about Y)
$\text{lift} < 1$	**Negative** association

Here $\text{lift}(A \to B) = 1.07 > 1$, so A and B are **positively associated**. Viewers who watched A went on to watch B slightly more often than a random viewer would.



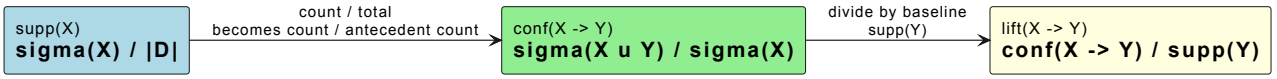
The purple bar is a little taller than the orange one. Knowing that a viewer watched A lifts the chance of B from 0.75 to 0.80, and $0.80/0.75 \approx 1.07$.

> **Note for Review Analogy:** suppose 75% of people in a town already own an umbrella. If 80% of people who bought a raincoat also own one, the raincoat tells you very little, because the town was already umbrella-heavy. Lift measures exactly this gap between the rule's probability and the background rate.
>
> **Pro-Tip:** lift is **symmetric** for a single pair, but confidence is not. Check: $\text{lift}(B \to A) = \dfrac{4/6}{5/8} = \dfrac{0.667}{0.625} \approx 1.07$, the same as $\text{lift}(A \to B)$. Exams like to test this.

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9. Summary of Terms

Term	Meaning
Item	$A, B, C, \ldots$ The fundamental unit or object (a product, a piece of content)
Transaction	One viewing session (or one purchase, one website visit). The collection of items observed together
Itemset	Any set of items, e.g. $\{A, B\}$, $\{A\}$, $\{B, D\}$, $\ldots$
Antecedent	Left-hand side of a rule. In $A \to B$ it is A
Consequent	Right-hand side of a rule. In $A \to B$ it is B
Support	Fraction of sessions containing the itemset, e.g. $\text{supp}(\{B, C\})$. It is $\sigma(X)/\lvert D\rvert$
Confidence	Fraction of sessions containing the antecedent that also contain the consequent, e.g. for $A \to B$
Lift	$\text{conf}(X \to Y)/\text{supp}(Y)$. Compares the rule against the baseline of 1



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10. The Goal of Association Rule Mining

Everything above was defined by *picking* a rule ($A \to B$) and computing its numbers. In practice we are never given a rule. We are given only the database:

$$D = \{T_1, T_2, T_3, \ldots\}$$

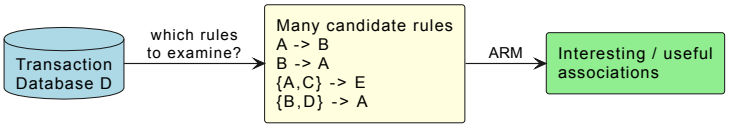
The universe is $I = \{A, B, C, D, E\}$ and nothing more. We do not know whether to examine

$$A \to B, \qquad B \to A, \qquad \{A, C\} \to E, \qquad \{B, D\} \to A, \quad \ldots$$

or whether such a rule even exists, how strong it is, or whether it makes sense.

> **Goal of ARM (Association Rule Mining):** Given the transaction database D, find the **interesting / useful associations**.

We want to know what is associated with what, which associations are positive or negative, and which items tend to come together. We also want to discard what is not useful for the customers or viewers.



11. Combinatorial Space Explosion

The main challenge is the **combinatorial space explosion**. Any item can associate with any other item. Two items can associate with another one, or three items with one, and so on. The number of combinations grows enormously.

11.1 Counting Itemsets

If there are m items, the number of possible (non-empty) itemsets is

$$2^m - 1$$

Why: each item is either in the itemset or not, which gives 2^m choices. We remove the empty set, so we get $2^m - 1$.

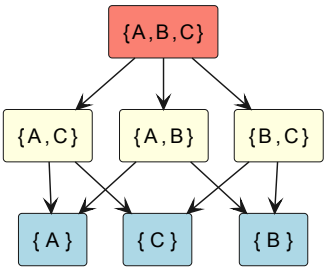
11.2 Worked Example: $m = 3$

Items $\{A, B, C\}$:

$$2^3 - 1 = 8 - 1 = 7$$

The seven itemsets are:

Size	Itemsets
1	$\{A\}, \{B\}, \{C\}$
2	$\{A, B\}, \{A, C\}, \{B, C\}$
3	$\{A, B, C\}$



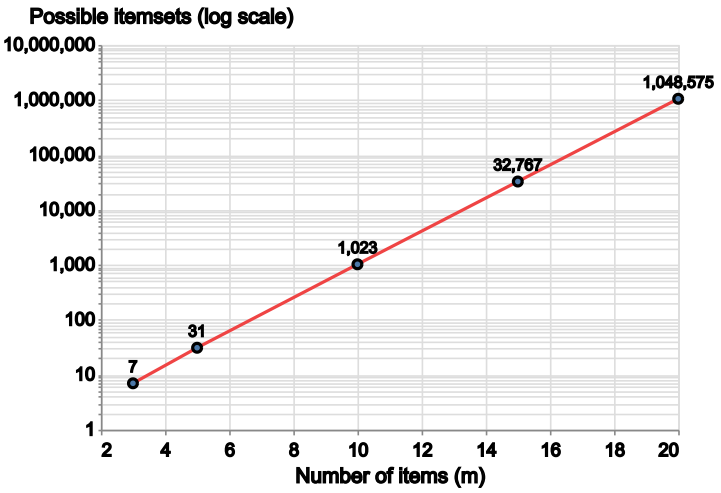
11.3 How Fast Does It Grow?

For $m = 20$ items:

$$2^{20} - 1 = 1{,}048{,}576 - 1 = 1{,}048{,}575$$

More than a million itemsets from only 20 items. A real store or platform might have $m = 100$ or more. This is why we call it an **explosion**: the growth is exponential.

The vertical axis is logarithmic, so the curve is still climbing steeply even though the picture looks tame. On a normal scale it would shoot straight up.



11.4 What We Do About It: Selectivity and Pruning

Association rules are built from itemsets. We therefore cannot afford to examine every combination of 20 (let alone 100) items. ARM needs a **selective search strategy**:

- Do **not** go through all combinations.
- **Prune** the search: throw away parts of the space that cannot be interesting, just as pruning in ML means discarding parts of a model (for example removing some layers of a neural network).
- Use **a priori knowledge** to decide what can be ignored.

The next lecture applies this idea to a small worked example, so we can see how much we gain by being selective and how much work we would have wasted without it.

> **Note for Review** Take-aways for this lecture: (1) know each definition precisely, (2) know what to count and what to divide by for support, confidence and lift, and (3) understand *why* ARM in its naive form is a problem, which is the combinatorial space explosion.

12. Practice Questions

Use the transaction table from Section 2 where data is needed.

Question 1: Rules that look strong but aren't. Suppose item Z is added to the database and appears in *every* transaction. Compute $\text{conf}(A \to Z)$ and $\text{lift}(A \to Z)$. A colleague proposes promoting Z to every viewer of A because the confidence is "perfect". Use your numbers to explain why this rule carries no useful information, and say which of the two measures exposes the problem.

Question 2: Direction matters, or does it? For the given data, compute $\text{conf}(A \to D)$ and $\text{conf}(D \to A)$, then $\text{lift}(A \to D)$ and $\text{lift}(D \to A)$. The two confidences differ. Do the two lifts differ? Explain why, using the numerators and denominators of each formula. Then state whether A and D are positively or negatively associated, and what that means in viewing behaviour.

Question 3: A neutral pair. Compute $\text{lift}(D \to E)$. The rule has a confidence of only 25%, which may sound like a "weak" rule. Does a low confidence necessarily mean a negative association? Justify your answer with the baseline $\text{supp}(E)$, and explain what a lift of exactly 1 tells a marketing team about promoting E to viewers of D.

Question 4: Budgeting the search. A platform has $m = 30$ pieces of content. (a) How many non-empty itemsets exist in total? (b) A data scientist decides, based on prior knowledge, that only itemsets of size at most 3 can be useful. How many itemsets must now be examined? (c) Compare the two numbers and describe, in your own words, what kind of "a priori knowledge" could justify such a cut-off and what risk it carries.

Question 5: Sample versus population, and causality. An analyst observes only these 8 sessions from a service with millions of viewers, reports "$\text{supp}(A, B) = 50\%$ for our whole user base", and then recommends *removing* content A from the catalogue to "test whether it drives people to watch B". Identify at least two distinct flaws in this reasoning. One should concern how far a number computed on 8 sessions can be trusted as a population-level statement. The other should concern what confidence and lift actually do (and do not) tell us about cause and effect.